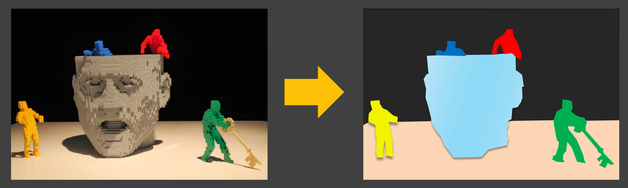
CutOut

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Motivation and Goals



- Image Segmentation is the process of partitioning an image into several different segments that compose it.
- Our Mission: implement a scribble-based algorithm for extracting object from natural photos and pasting them seamlessly into a different background under the constraints of a mobile device's computational power.

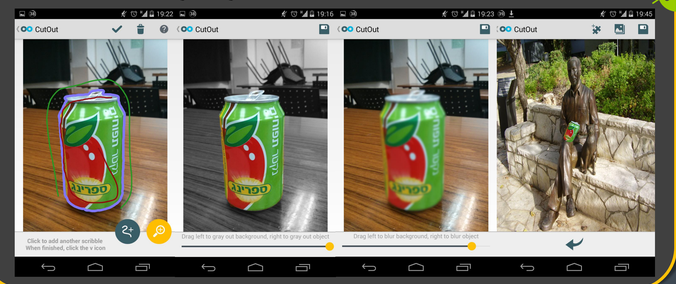
The Algorithm

- The algorithm first calculates the designated weight functions, using the scribbles.
- Distances are then calculated on the entire image and used to derive the final result.



The Application

- OpenCV was used to implement the algorithm on a personal computer.
  - C++ algorithm was ported to Android using the NDK.
- Android's SDK was used to wrap the algorithm in a friendly user interface that anybody can use.



The Main Idea

- The user provides us with two scribbles
  - The pixels in the inner scribble belong to the object.
  - The pixels in the outer scribble belong to the background.
- Propagate that information to the rest of the image pixels using a designated weight distance function

$$d(s_1,s_2) := \min_{C_{s_1,s_2}} \int_0^1 \left| W \cdot \dot{C}_{s_1,s_2}(p) \right| dp \qquad D_l(x) = \min_{s \in \Omega_l}(s,x), \quad l \in \{\mathcal{F},\mathcal{B}\}$$

- Define weights using the **P**robability **D**ensity **F**unctions of both scribbles

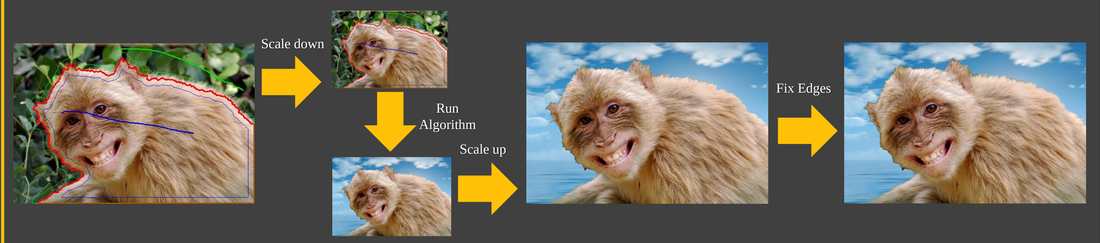
$$P_{\mathcal{F}}(x) = \frac{Pr(x|\mathcal{F})}{Pr(x|\mathcal{F}) + Pr(x|\mathcal{B})} \qquad W = \nabla P_{\mathcal{F}}(x)$$

- Define the object as the pixels that are closer to the inner scribble

$$x \in \mathcal{O} \Leftrightarrow D_{\mathcal{F}}(x) < D_{\mathcal{B}}(x)$$

Improving runtime

- Running Dijkstra's algorithm on a full-sized image can be computationally heavy.
- Solution: run the main algorithm on a resized image, add another refinement step for fixing the edges.

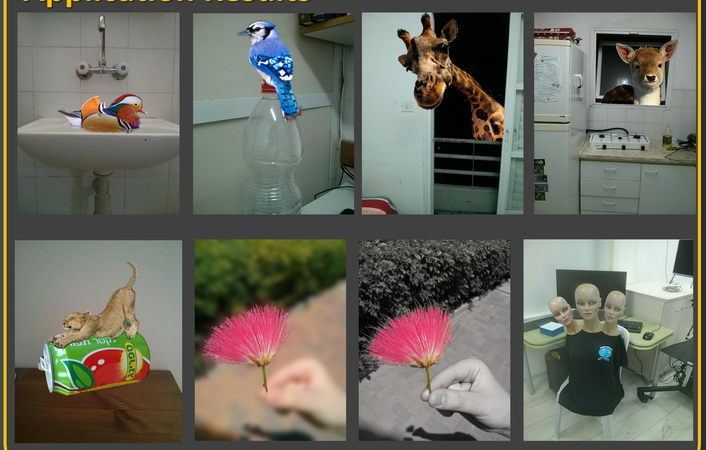


- Edges are fixed using both blurring and dedicated alpha calculation.

Matting Results



Application Results



References

- "A Geodesic Framework for Fast Interactive Image and Video Segmentation and Matting", by Xue Bai and Guillermo Sapiro
- "Image Editing Using Level Set Trees", by Anastasia Dubrovina, Rom Hershkovitz and Ron Kimmel

GET IT ON Google play

What users say

- Works like magic! ★★★★★
- Molto bello ★★★★
- Abracadabra, really works! Just what I needed! ★★★★★
- Solid App… The process is unique but effective Kudos to the Developer. ★★★★